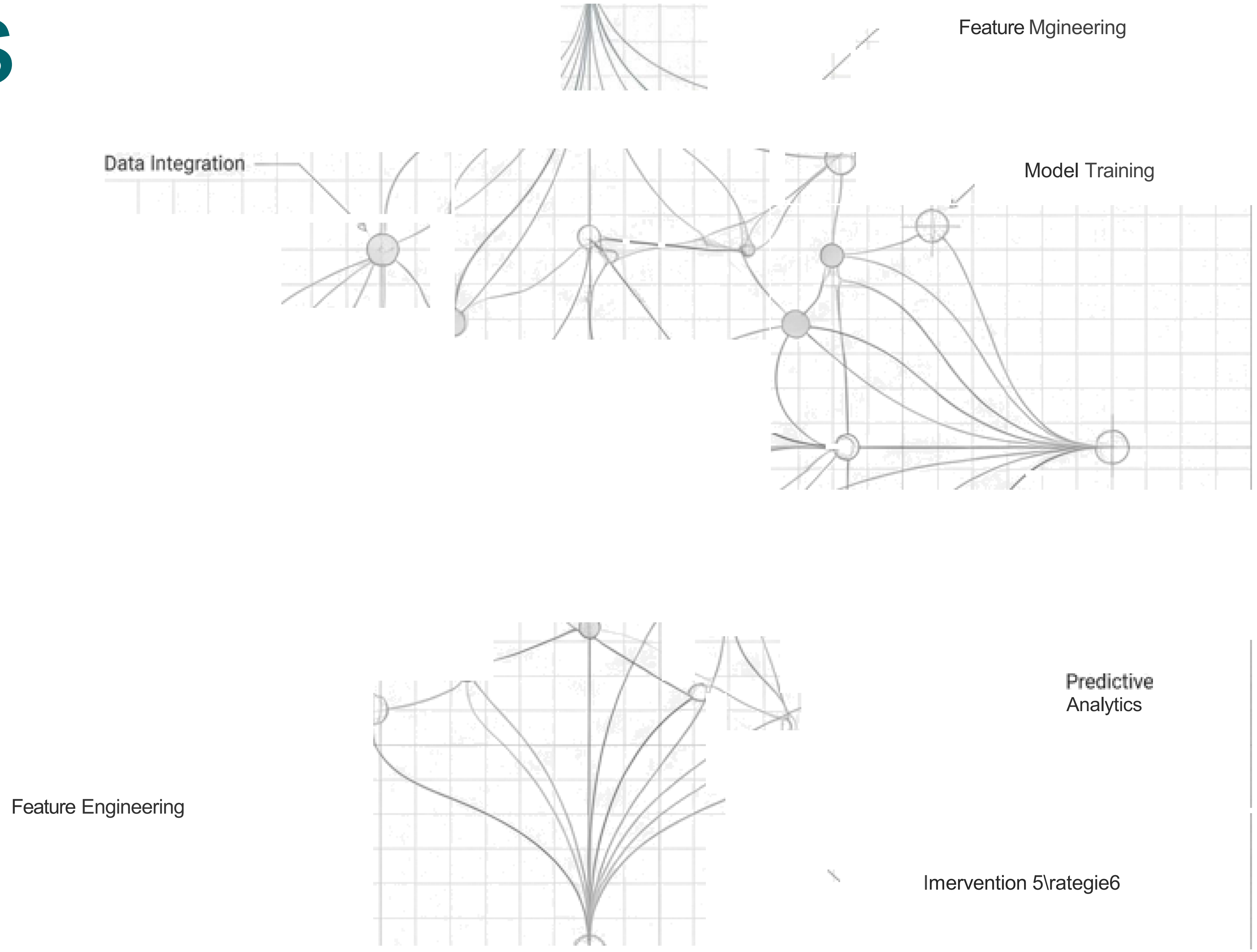
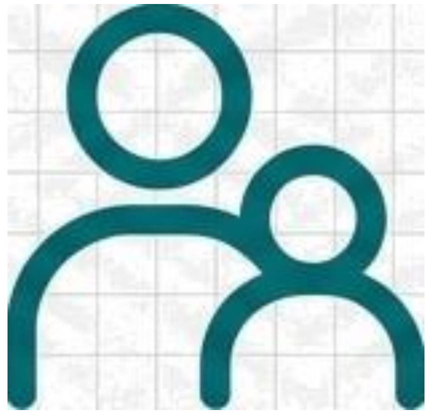
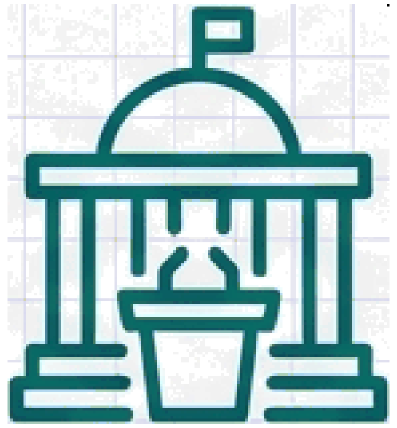
Cracking the Code of
of Student Success

An End-to-End Machine Learning Project



Why Predict Student Performance?





For Educators

To identify students who may need additional support.

For Parents

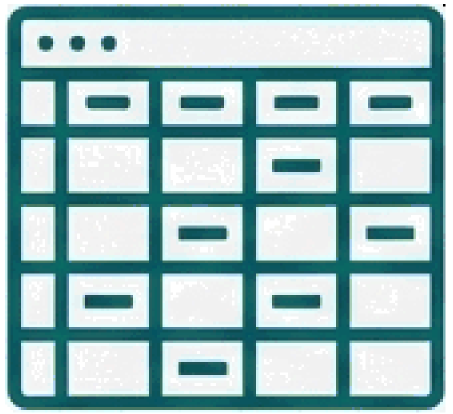
To understand key influencing factors.

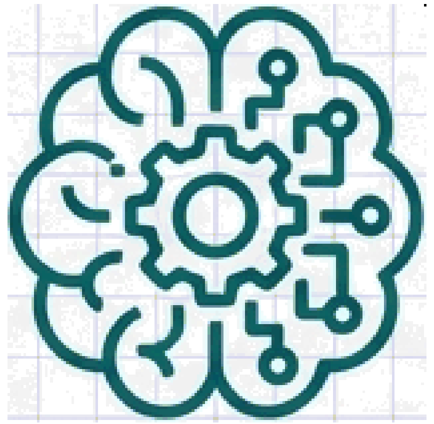
For Policymakers

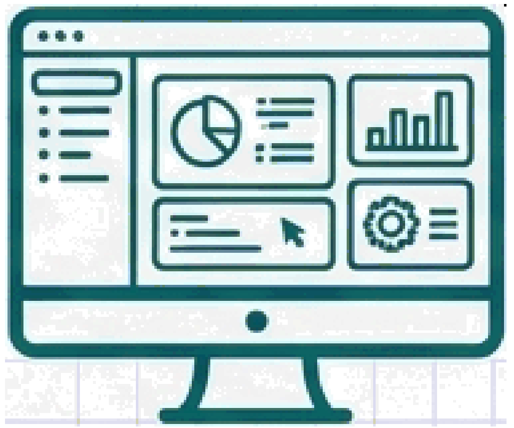
To help tailor effective educational strategies.

“Understanding the factors that contribute to a student’s academic performance is crucial.”

The Vision: From Raw Data to a Live Application







1. Analyze

Start with a real-world dataset.

2. Build

Develop a predictive ML model.

3. Deploy

Deliver insights through an interactive web app.

This project demonstrates the complete, end-to-end process of developing and deploying a machine learning solution.

The Raw Material: The Students Performance in Exams Dataset

e etails

Source: Kaggle

Content: A rich mix of student information.

Featu es

Gender

Race/Ethnicity

Parental Level of Education

Lunch Type

Test Preparation Course

Reading & Writing Scores

Target: Math Score

The Architectural Blueprint

src/	The Core Logic
components/	Reusable modules (Data Ingestion, Model Training)
pipeline/	The end-to-end workflow orchestrator
notebook/	Exploratory Data Analysis
templates/ & static/	The Web Application Front-End
artifacts/	The trained model output

A Modern Data Science Toolkit

Data Manipulation & Analysis

NumPy matplotlib

Web Framework & Deployment

Flask

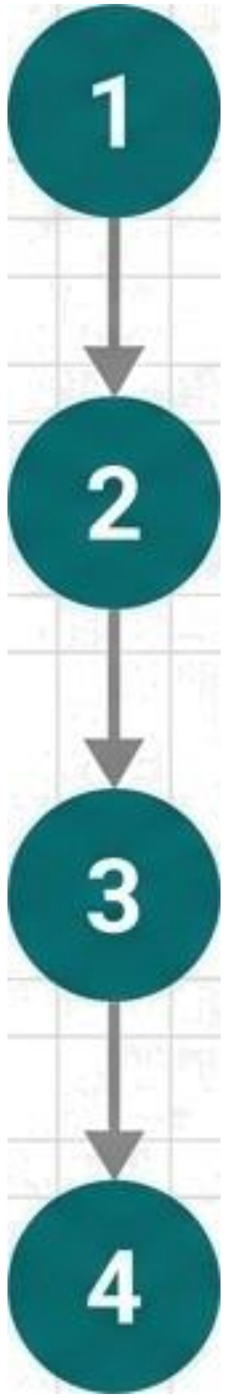
Machine Learning & Modeling

scikit learn XGBoost

Model Serialization

dill

Engineering the Prediction Pipeline



1. Data Ingestion

Read the raw data.

2. Data Transformation

Handle categorical features and scale numerical data.

3. Model Training

Split data into training/testing sets and train the model.

4. Artifact Generation

Save the trained model and preprocessor for inference.

The Results: 85% Accuracy and Key Insights

85%

Achieved accuracy in predicting student math scores.

What Factors Matter Most?

- Parental Level of Education
- Completion of Test Preparation Course
- Lunch Type

The Fina! Product: A Student Performance Predictor

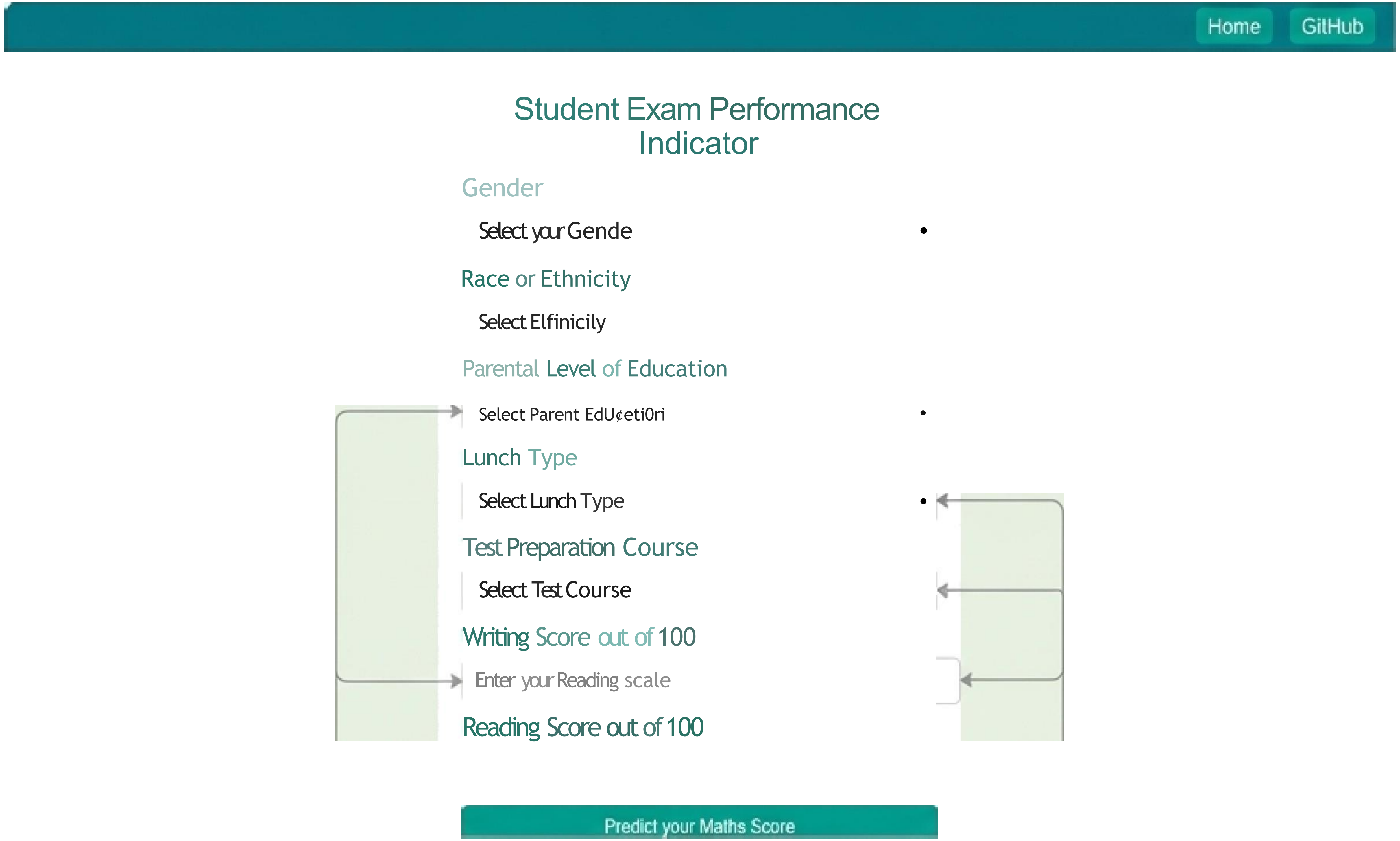
Home GitHub

Predicting Student Performance using Machine Learning

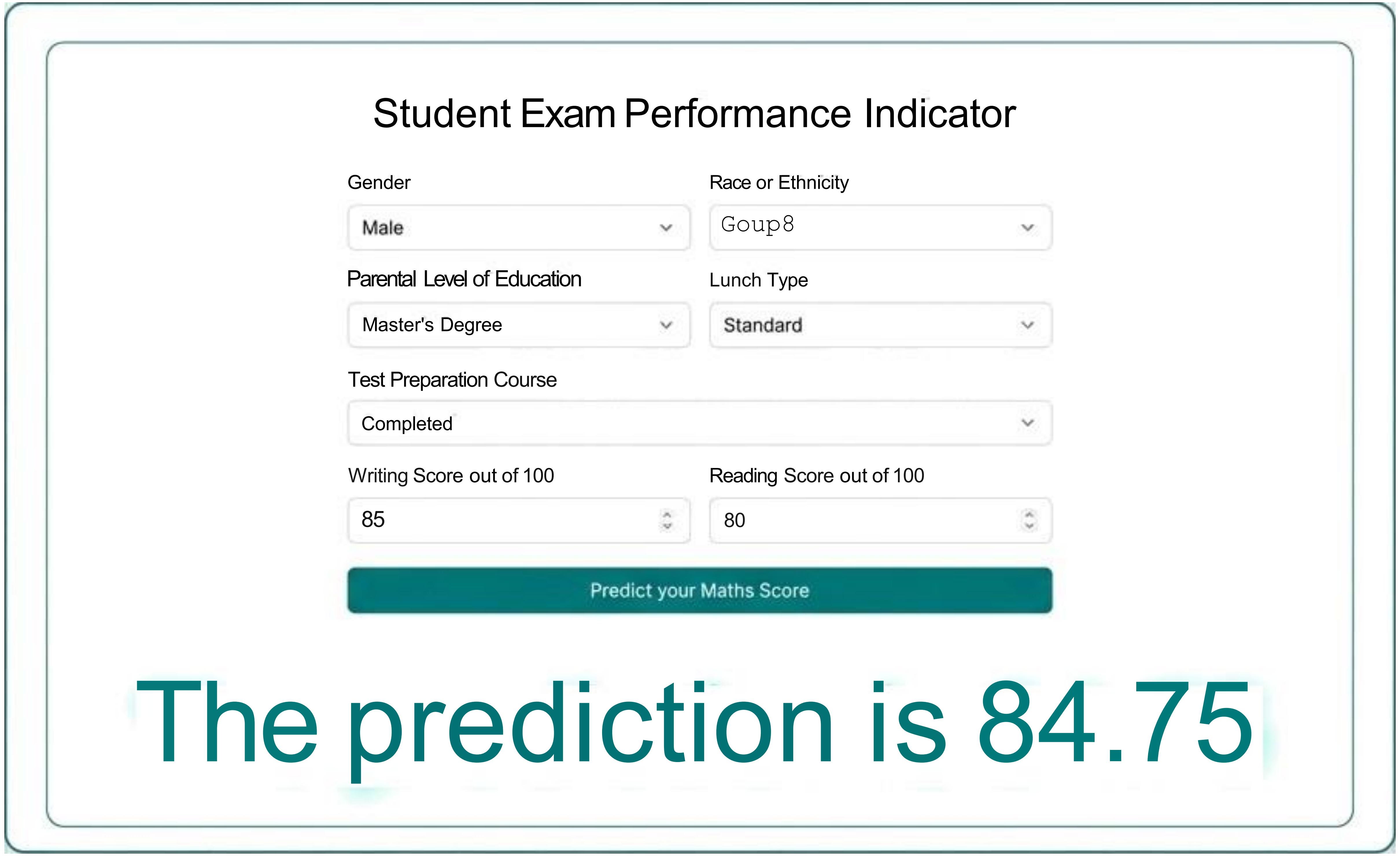
This project endeavours to develop a machine learning model capable of predicting student performance in mathematics based on a range of factors.

GitHub Repository Web App

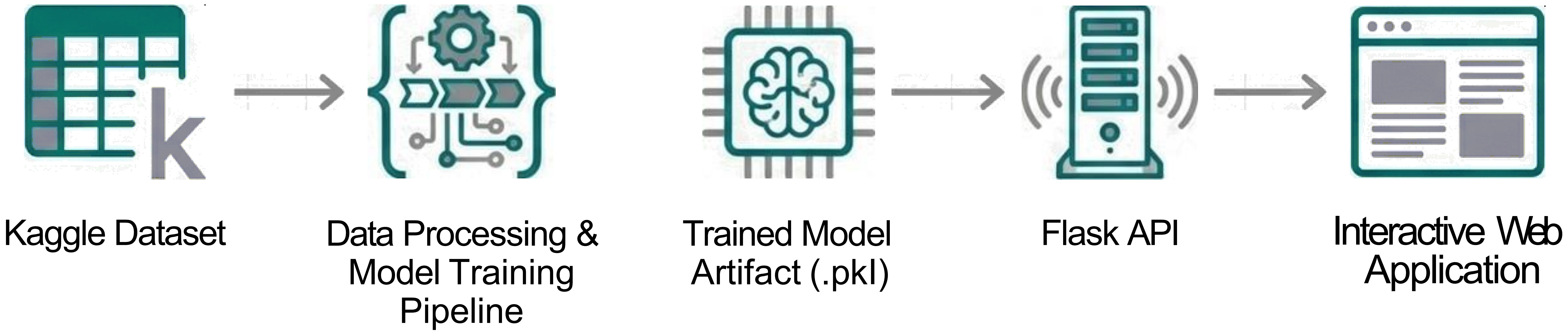
Putting the Model to Work: The Input Interface



From Input to Instant Insight



The Complete End-to-End Workflow



More Than a Model: A Demonstration of Process

"The primary goal of this project is to demonstrate the end-to-end process of developing a machine learning model and provide insights into the factors influencing student performance."

- ,x" Showcased a structured approach to solving a problem with ML.
- ,x' Translated complex data into an accessible, interactive tool.
- ,z' Built a foundation for a tool that could offer real-world educational benefits.

The Blueprint is Open

Explore the repository, run the code, and build on this work.

View on GitHub:-
github.com/jamilmrt/Predicting-Student-Performance-Using-Machine-Learning